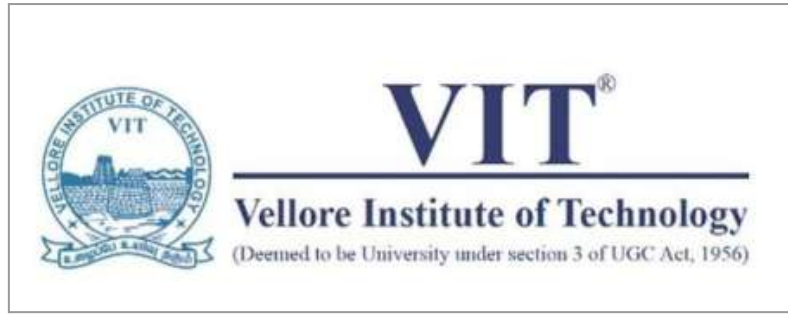




School of Computer Science and Engineering (SCOPE)

Fall Semester 2025-26

CBS3005 - Cloud, Microservices and Applications

LAB ASSESSMENT 4

Name: Riya Autade

Registration No.: 22BBS0079

Q1) A company wants to deploy a secure, scalable, and highly available web application on AWS for global users. Perform the following tasks in AWS and submit screenshots of each step as evidence:

- (i) Launch multiple EC2 instances (web servers) and configure them in different Availability Zones.
- (ii) Create an Application Load Balancer (ALB) to distribute traffic across these instances.
- (iii) Configure health checks so that faulty instances are automatically removed from load balancing.
- (iv) Enable Auto Scaling to add/remove instances based on traffic demand.
- (v) Configure path-based routing: /auth requests go to the authentication service, /order requests go to the order processing service. Integrate the Load Balancer with Route 53 so that global users are routed to the nearest AWS region.

STEPS:

AWS Assignment: Deploy Secure, Scalable, Highly Available Web App

Step 1: Launch EC2 Instances in Different Availability Zones

Create 3 EC2 instances, each serving a different service ([home](#), [auth](#), [order](#)).

Common Settings:

- AMI: Amazon Linux 2
 - Instance Type: t2.micro
 - Security Group: Allow SSH (22) and HTTP (80)
 - VPC: Same for all three instances but different subnets in different AZs.
-

Instance 1: Homepage

- Availability Zone: us-east-1b (select subnet in this AZ)
- User Data script:

```
bash
#!/bin/bash
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd

cat <<EOF > /var/www/html/index.html
<!DOCTYPE html>
<html>
```

```
<head><title>My AWS Web App</title></head>
<body>
  <h1>Welcome to My AWS Web App</h1>
  <p>Homepage served here!</p>
  <nav>
    <a href="/auth">Authentication Service</a> |
    <a href="/order">Order Service</a>
  </nav>
</body>
</html>
EOF
```

Instance 2: Authentication Service

- Availability Zone: us-east-1a
- User Data script:

```
bash
#!/bin/bash
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd

cat <<EOF > /var/www/html/index.html
<!DOCTYPE html>
<html>
<head><title>Authentication Service</title></head>
<body>
  <h1>Authentication Service</h1>
  <p>This is the /auth path</p>
</body>
</html>
EOF
```

Instance 3: Order Service

- Availability Zone: us-east-1c
- User Data script:

```
bash
#!/bin/bash
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd

cat <<EOF > /var/www/html/index.html
<!DOCTYPE html>
<html>
<head><title>Order Service</title></head>
<body>
  <h1>Order Service</h1>
  <p>This is the /order path</p>
</body>
</html>
EOF
```

Step 2: Create Target Groups

For each service, create a Target Group:

- Type: Instance
- Protocol: HTTP
- Port: 80
- Health Check Path: /

Attach the appropriate EC2 instance only to each target group:

- TG-Homepage → Homepage EC2 instance

- TG-Auth → Auth EC2 instance
 - TG-Order → Order EC2 instance
-

Step 3: Create the Application Load Balancer (ALB)

- Scheme: Internet-facing
 - Select all three subnets (corresponding to your AZs)
 - Security Group: Allow inbound HTTP (80)
 - Listener Protocol: HTTP port 80
 - Default Action: Forward to TG-Homepage target group
-

Step 4: Configure ALB Listener Rules (After creation)

- In EC2 → Load Balancers → Select your ALB → **Listeners and rules**
 - Click on Rules for HTTP:80 listener
 - Add rules:
 - If path is `/auth*`, forward to TG-Auth target group
 - If path is `/order*`, forward to TG-Order target group
 - Default rule forwards all other requests to TG-Homepage
-

Step 5: Configure Health Checks

For each Target Group:

- Protocol: HTTP
 - Port: 80
 - Path: /
 - Healthy threshold: 2
 - Unhealthy threshold: 5
 - Interval: 30 seconds (default)
 - Timeout: 5 seconds (default)
-

Step 6: Enable Auto Scaling (Repeat for each service)

1. Create a Launch Template for each service with service-specific User Data.
 2. Create an Auto Scaling Group with:
 - Min capacity: 1
 - Max capacity: 2
 - Network: Select subnets in appropriate AZs
 - Attach ALB's Target Group for this service
 3. This allows scaling instances up/down.
-

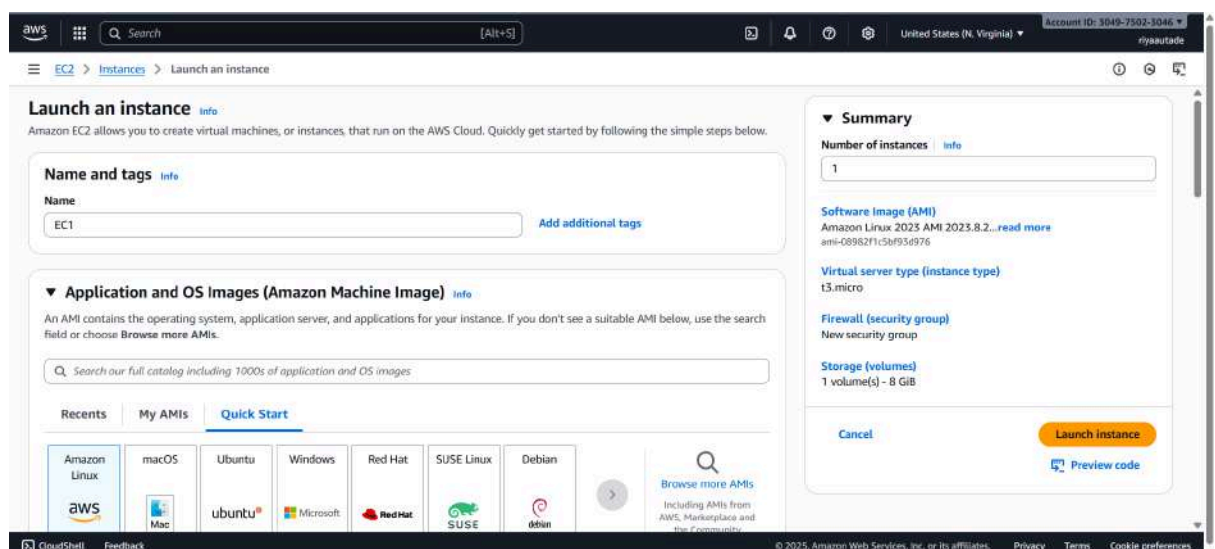
Step 7: Access Using ALB DNS Name (Without Domain)

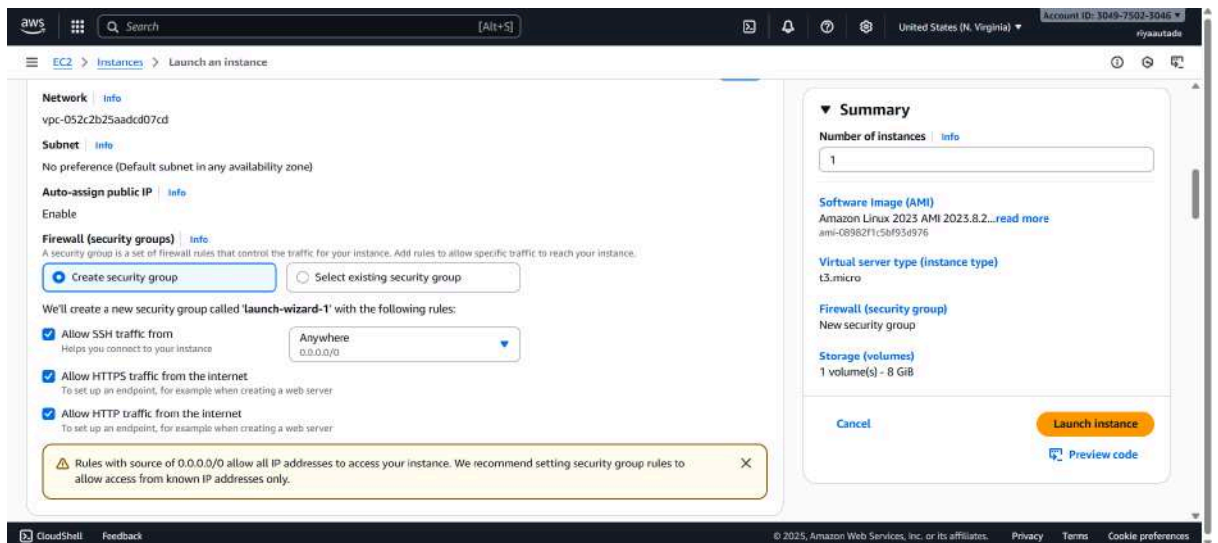
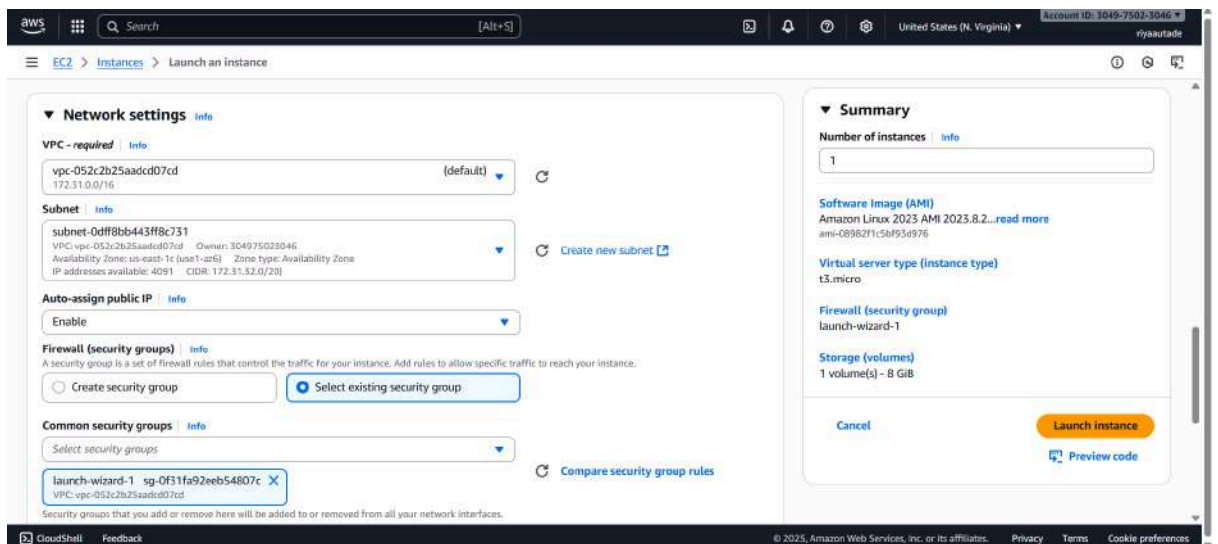
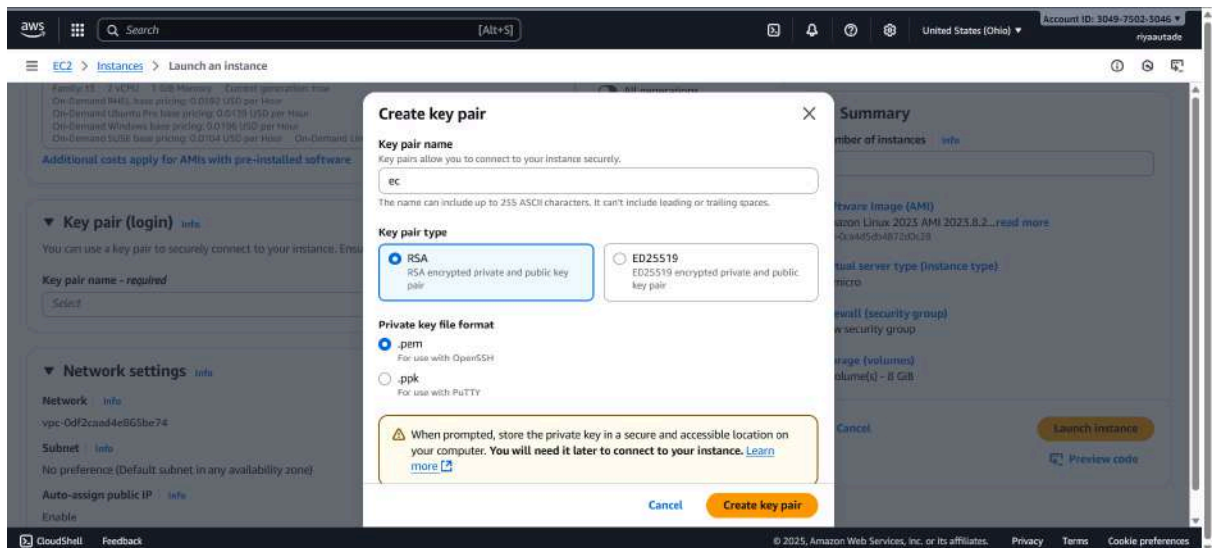
- Find your ALB's DNS name from EC2 → Load Balancers → select your ALB → Description tab
 - Use in browser to access services:
 - http://<ALB_DNS>/ → Homepage
 - http://<ALB_DNS>/auth → Auth Service
 - http://<ALB_DNS>/order → Order Service
-

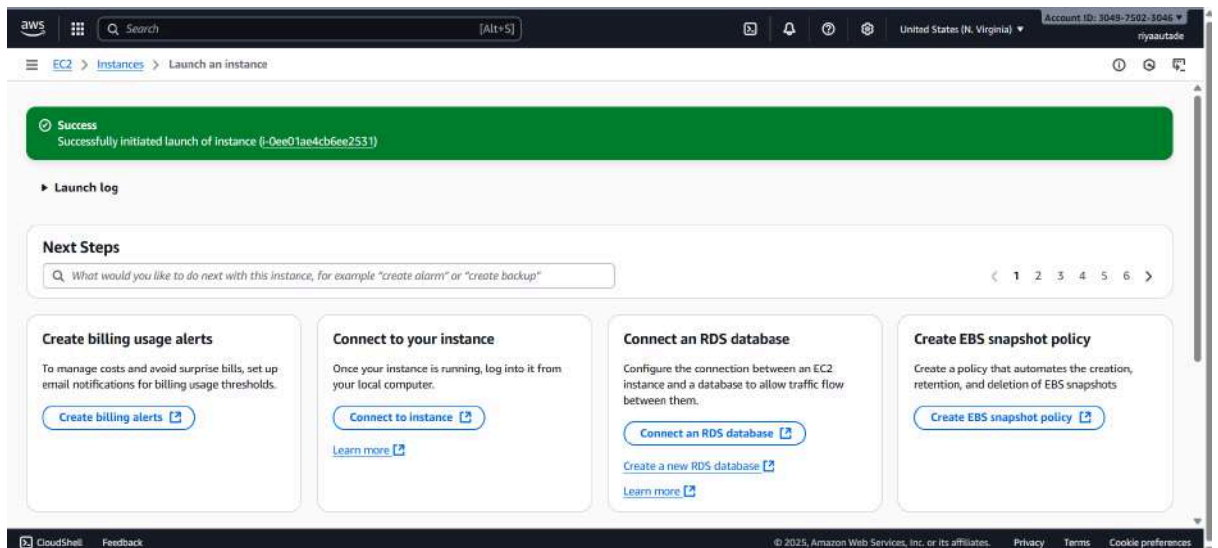
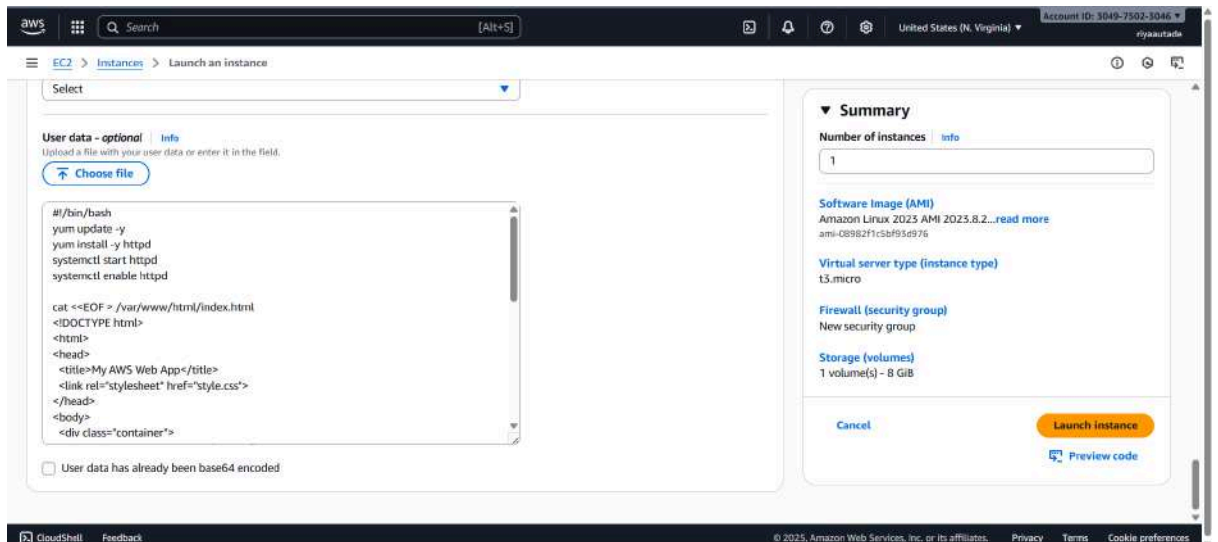
Optional Step 8: Route 53 Integration (If you have a domain)

- Create a Hosted Zone for your domain.
- Add an Alias A Record pointing to your ALB's DNS.
- Optionally configure Latency-based routing for global ALBs.
- This enables users to access via your friendly domain name.

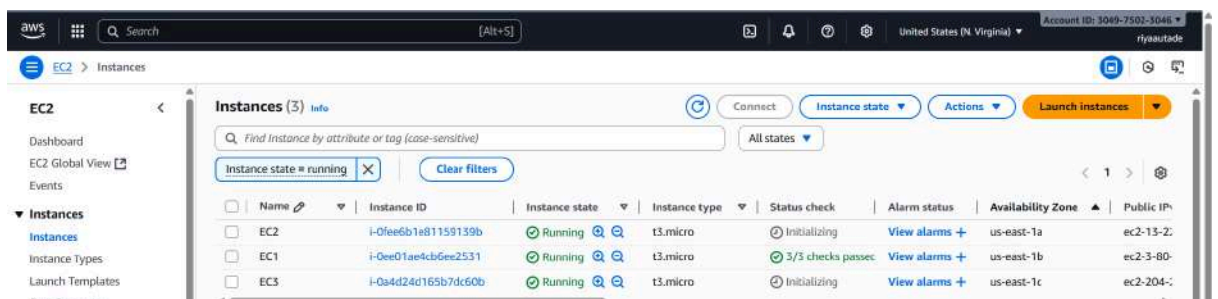
1. Launch 3 instances in different AZs:



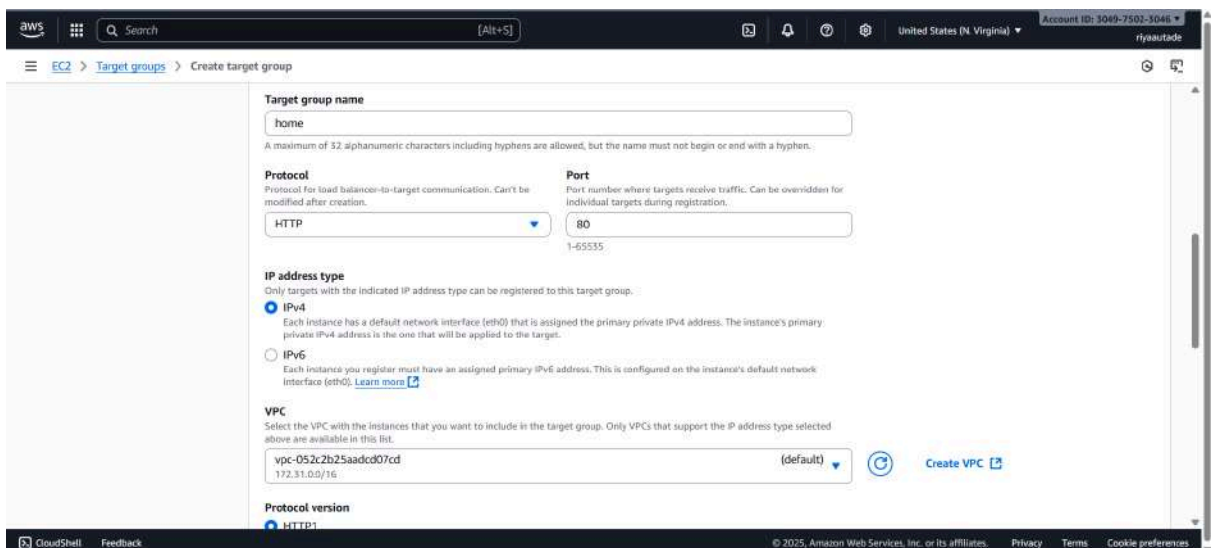
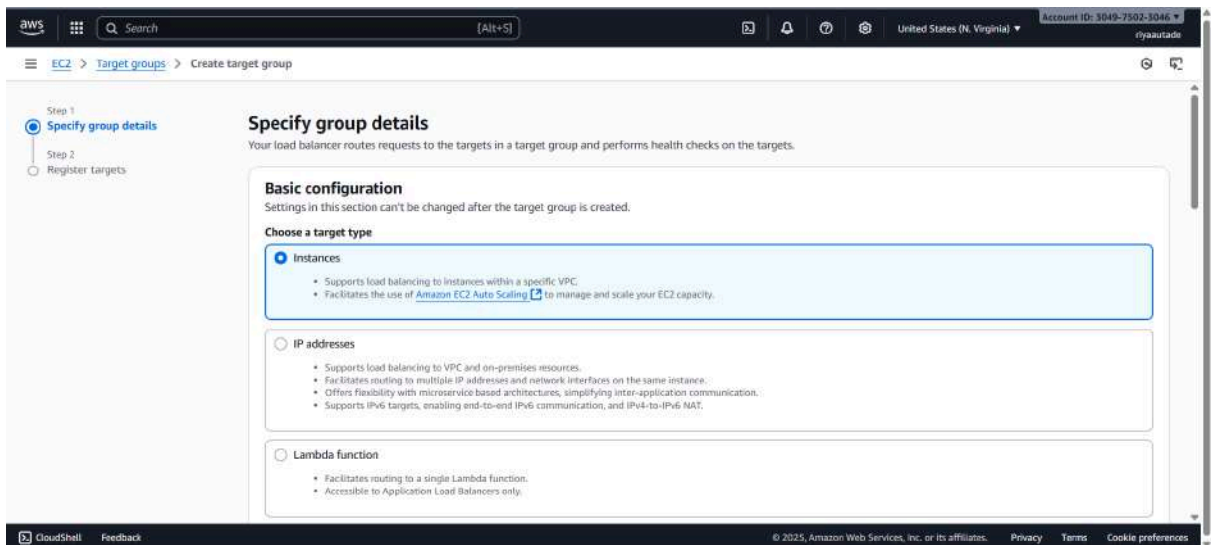
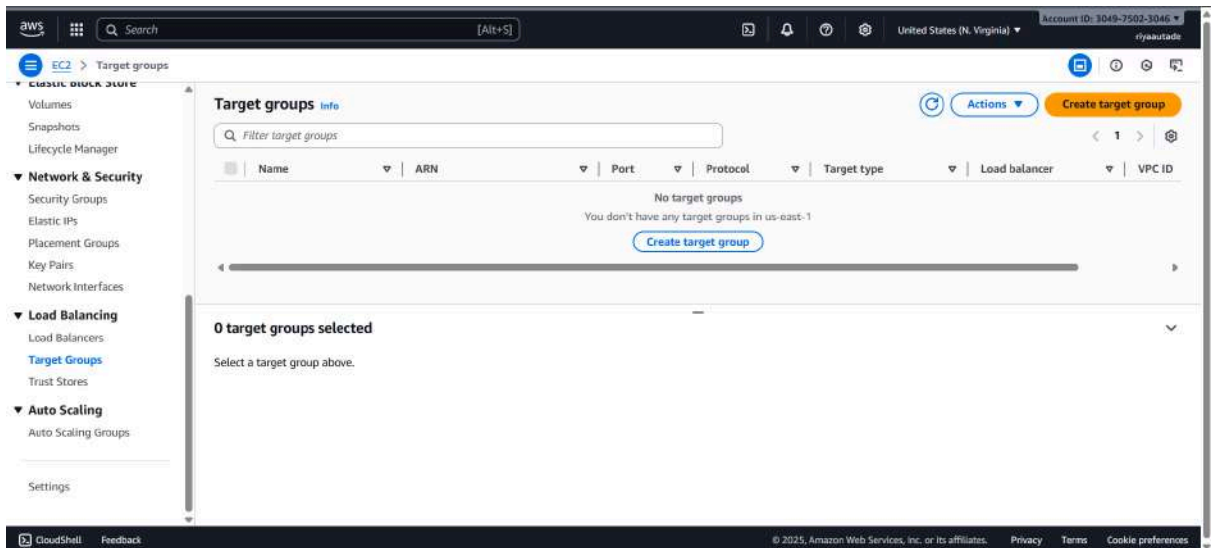




Us-east-1a
Us-east-1b
Us-east-1c



2. Create target groups:



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EC2 > Target groups > Create target group

Protocol version

☒ HTTP1

Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ HTTP2

Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ gRPC

Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Health checks

The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol

HTTP

Health check path

Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.

/

Up to 1024 characters allowed.

Advanced health check settings

CloudShell

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EC2 > Target groups > Create target group

Step 1
Specify group details

Step 2
Register targets

Register targets

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (1/3)

Filter instances

	Instance ID	Name	State	Security groups	Zone
☐	i-0a4d24d165b7dc60b	EC3	Running	launch-wizard-1	us-east-1c
☐	i-0fee6b1e81159139b	EC2	Running	launch-wizard-1	us-east-1a
☒	i-0ee01ae4cb6ee2531	EC1	Running	launch-wizard-1	us-east-1b

1 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

80

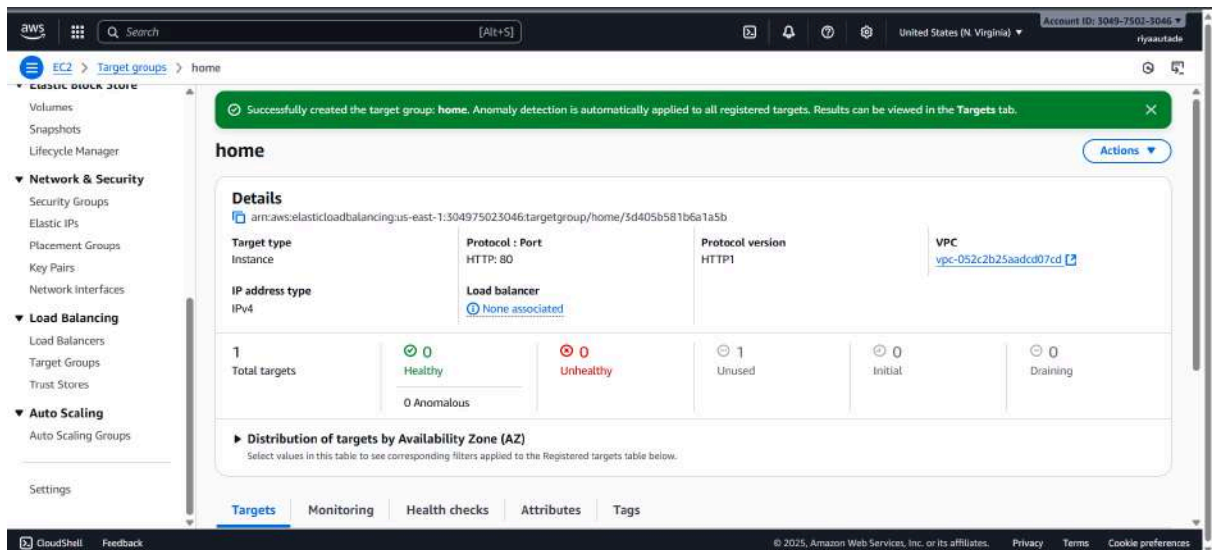
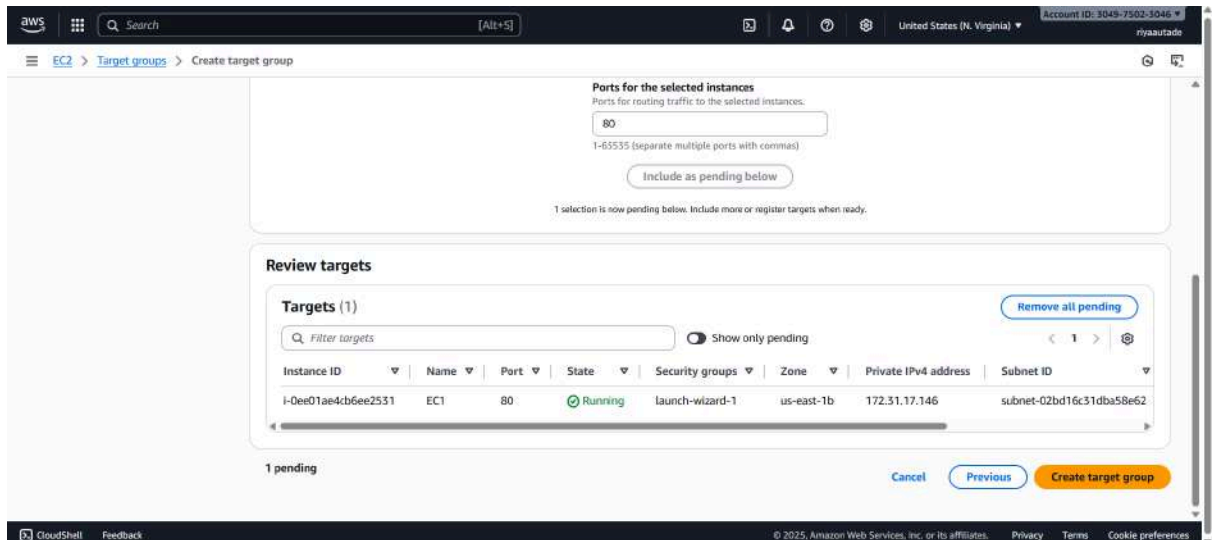
1-65535 (separate multiple ports with commas)

Include as pending below

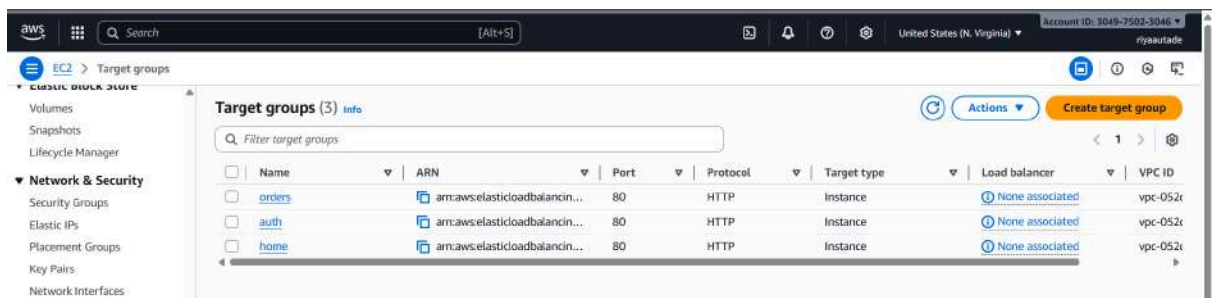
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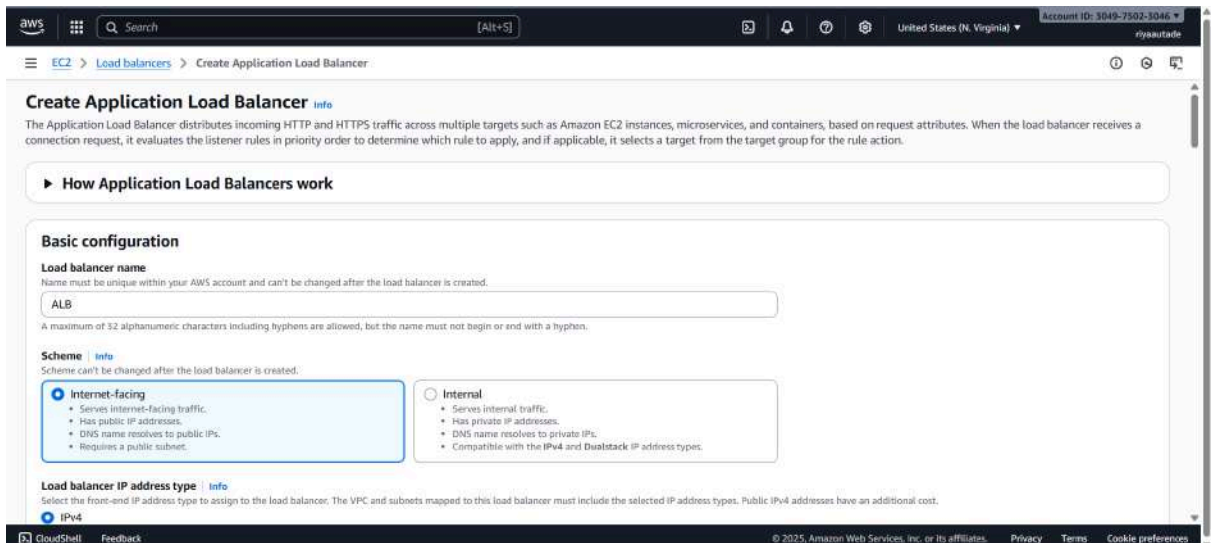
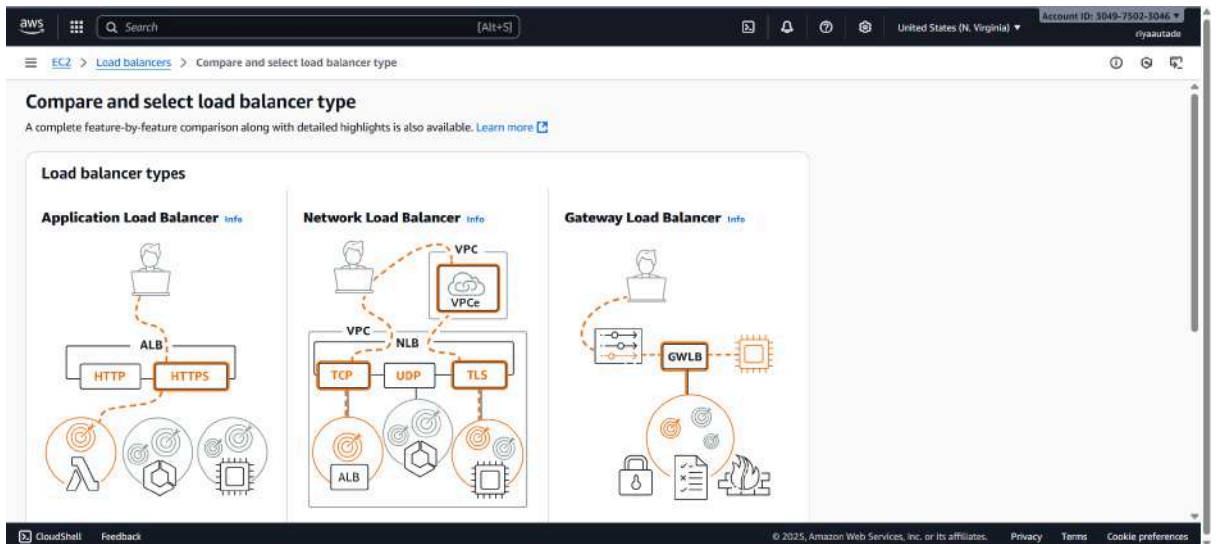
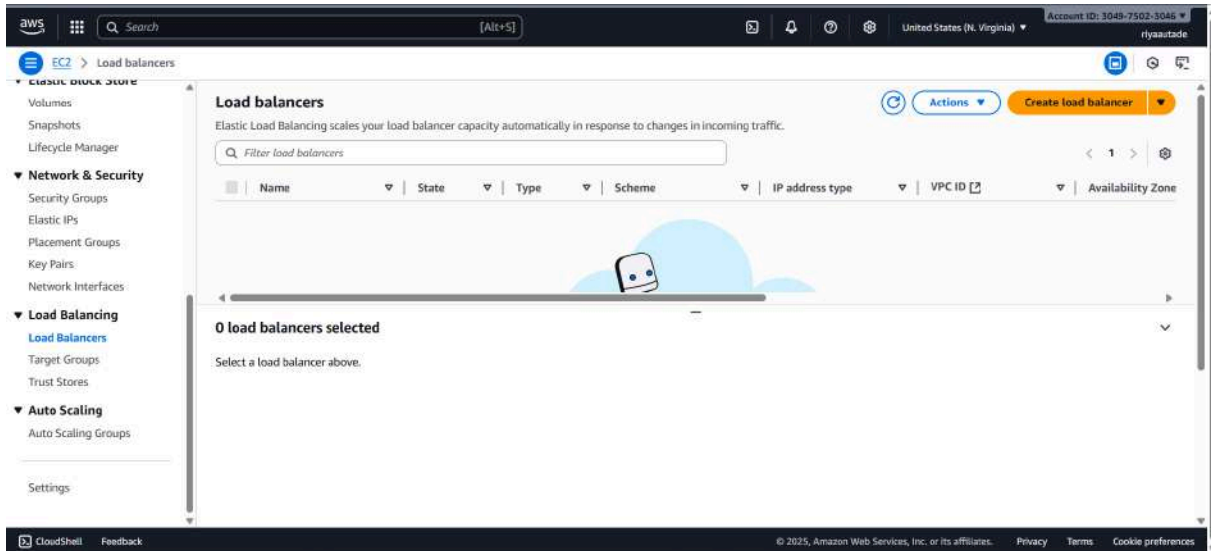
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Repeat



3. Create and Configure Application Load Balancer (ALB)



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EC2

Load balancers

Create Application Load Balancer

Network mapping

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-052c2b25aadcd07cd

{default}

Create VPC

IP pools

You can optionally choose to configure an IPAM pool as the preferred source for your load balancer IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ us-east-1a (use1-az2)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0dc025dd6af7da987

IPv4 subnet CIDR: 172.31.0.0/16

☒ us-east-1b (use1-az4)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-02bd15c31dba58e62

IPv4 subnet CIDR: 172.31.16.0/20

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EC2

Load balancers

Create Application Load Balancer

Security groups

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

launch-wizard-1

sg-0f31fa92web54807c VPC: vpc-052c2b25aadcd07cd

Listeners and routing

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80 Remove

Protocol HTTP **Port** 80

1-65535

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EC2

Load balancers

Create Application Load Balancer

Default action

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups ☐ Redirect to URL ☐ Return fixed response

Forward to target group

Choose a target group and specify routing weight or [create target group](#).

Target group	Weight	Percent
home Target type: Instance, IPv4 Target stickiness: Off	1 0-999	100%

[+ Add target group](#)

You can add up to 4 more target groups.

Target group stickiness

Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

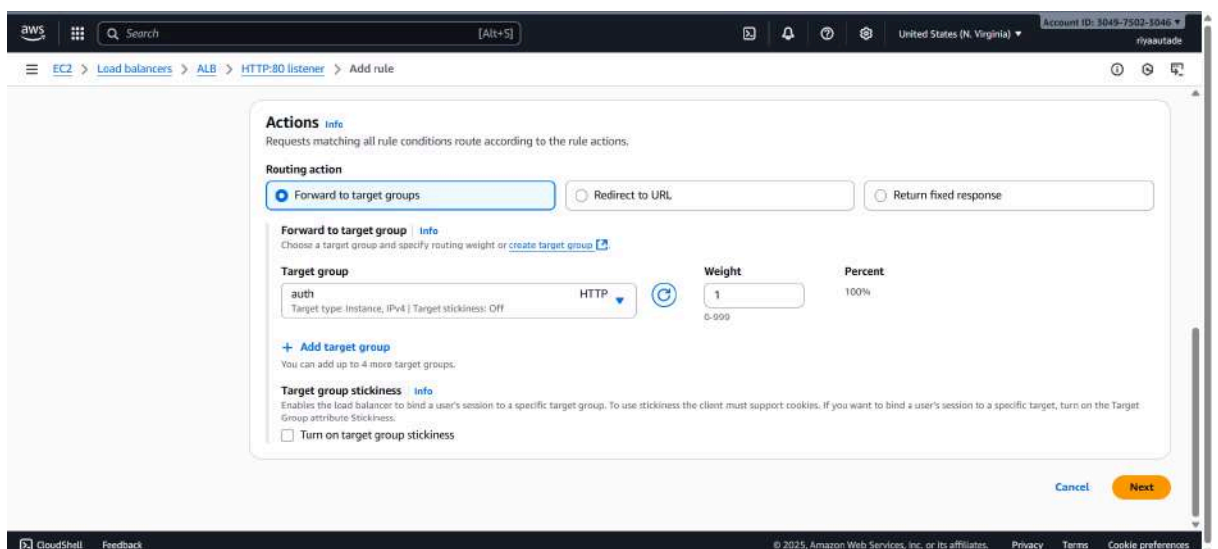
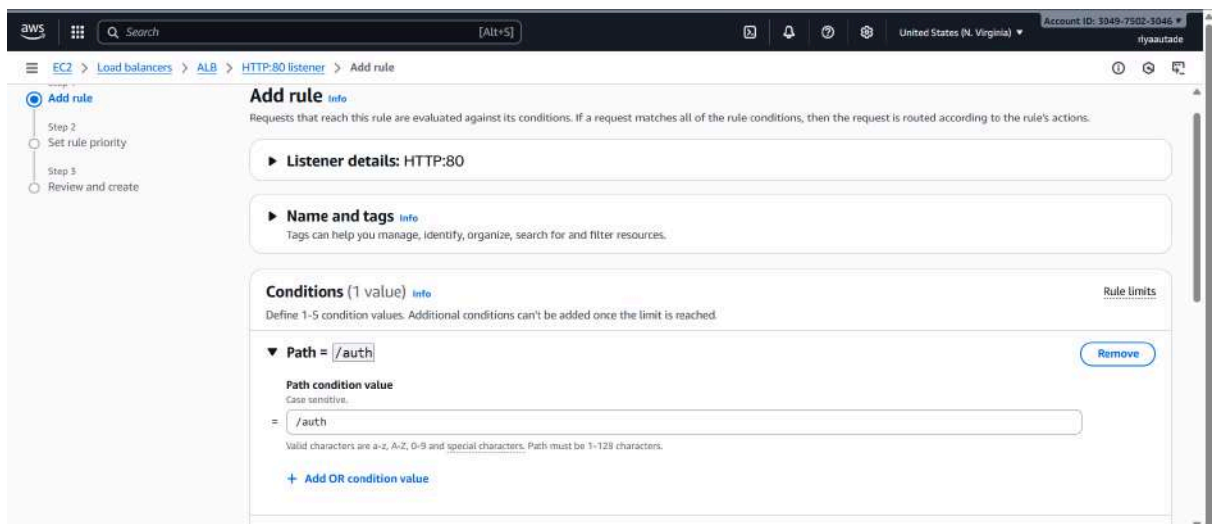
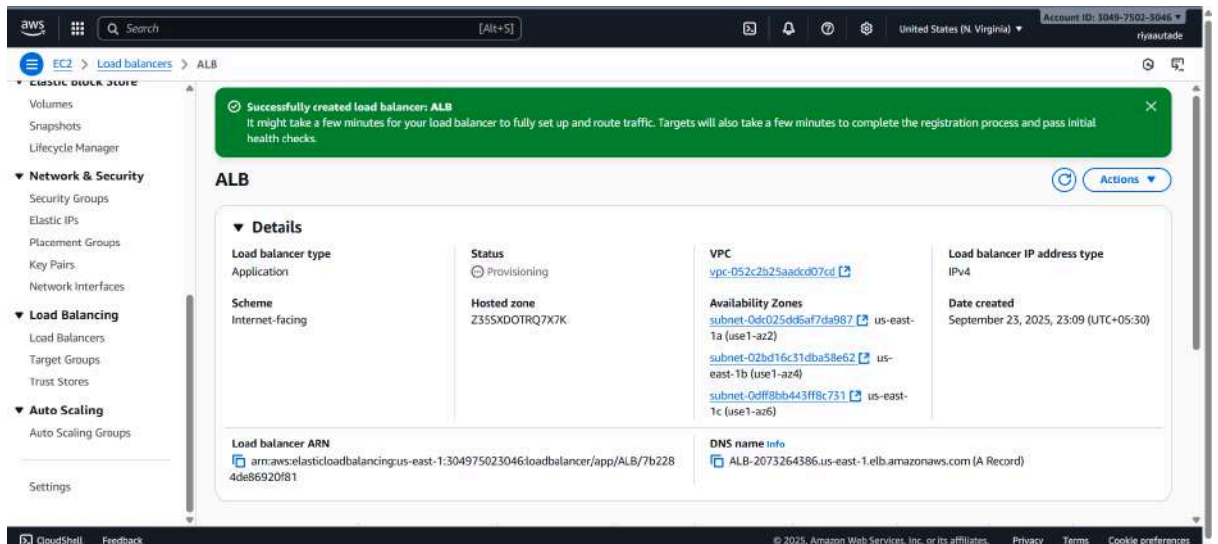
[Add listener tag](#)

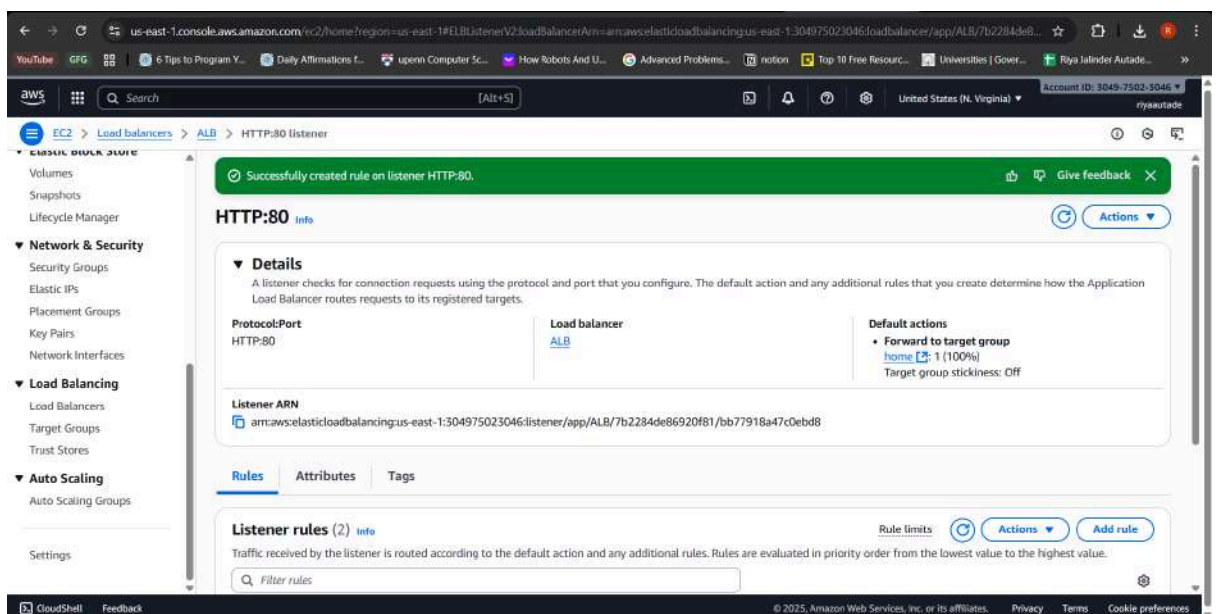
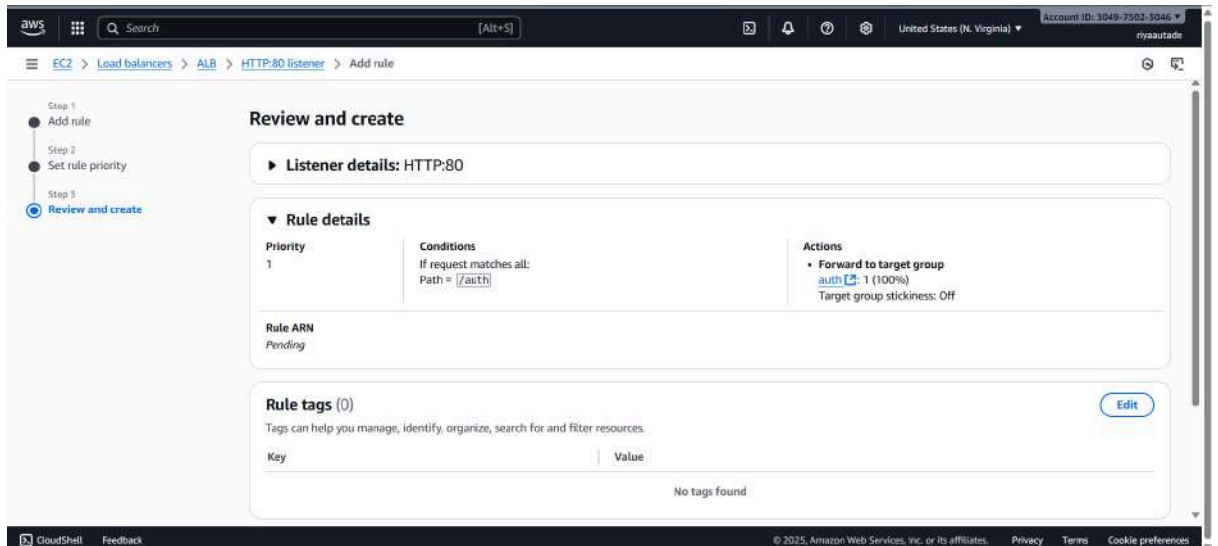
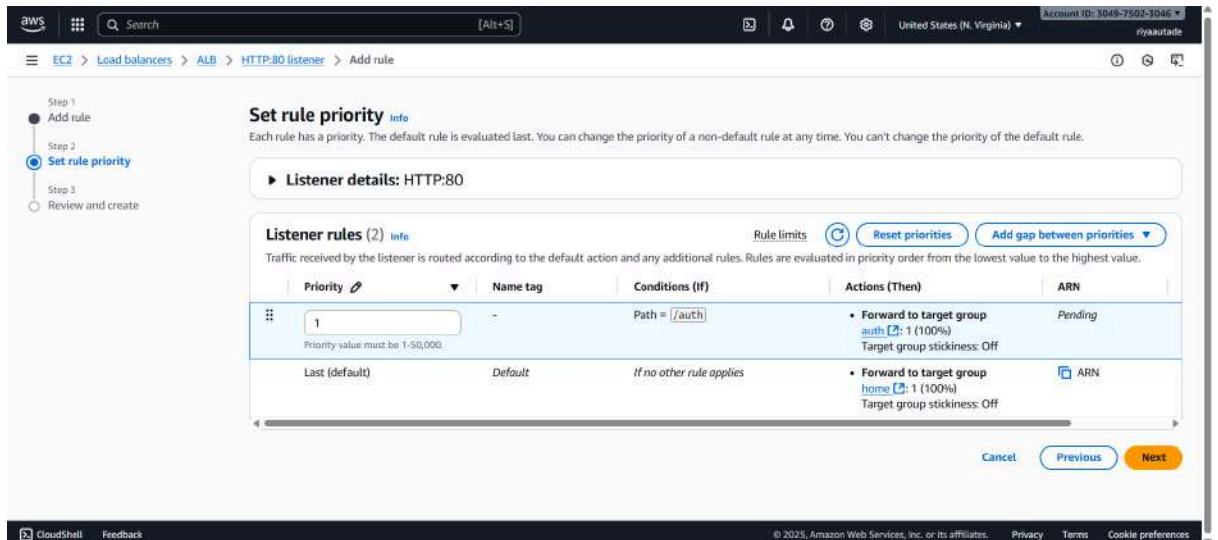
You can add up to 50 unique tags.

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Similarly for /orders

The screenshot shows the 'Listener rules' configuration page in the AWS Management Console. It displays three rules in a table:

Priority	Name tag	Conditions (If)	Actions (Then)	ARN	Actions
1	-	Path = /auth	Forward to target group auth : 1 (100%) Target group stickiness: Off		
2	-	Path = /order	Forward to target group orders : 1 (100%) Target group stickiness: Off		
Last (default)	Default	If no other rule applies	Forward to target group home : 1 (100%) Target group stickiness: Off		

4. Health checks

The screenshot shows the 'Health checks' configuration page for the 'home' target group. The 'Health check settings' section is expanded, showing the following configuration:

Setting	Value
Protocol	HTTP
Path	/
Port	Traffic port
Unhealthy threshold	2 consecutive health check failures
Timeout	5 seconds
Interval	30 seconds
Healthy threshold	5 consecutive health check successes
Success codes	200

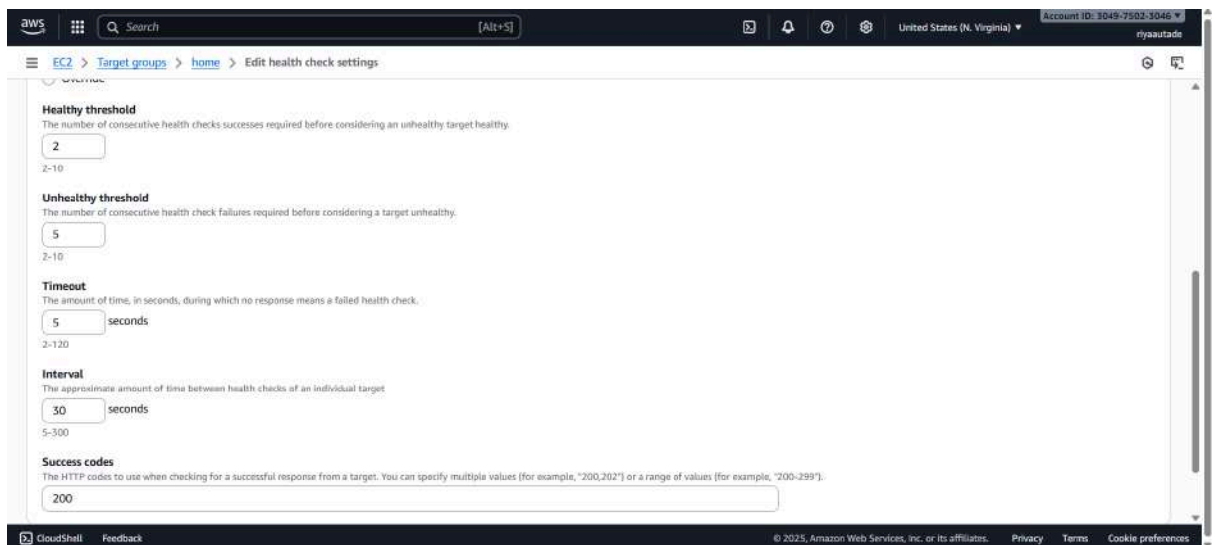
The screenshot shows the 'Edit health check settings' page. The 'Health check settings' section is expanded, showing the following configuration:

Health check protocol: HTTP

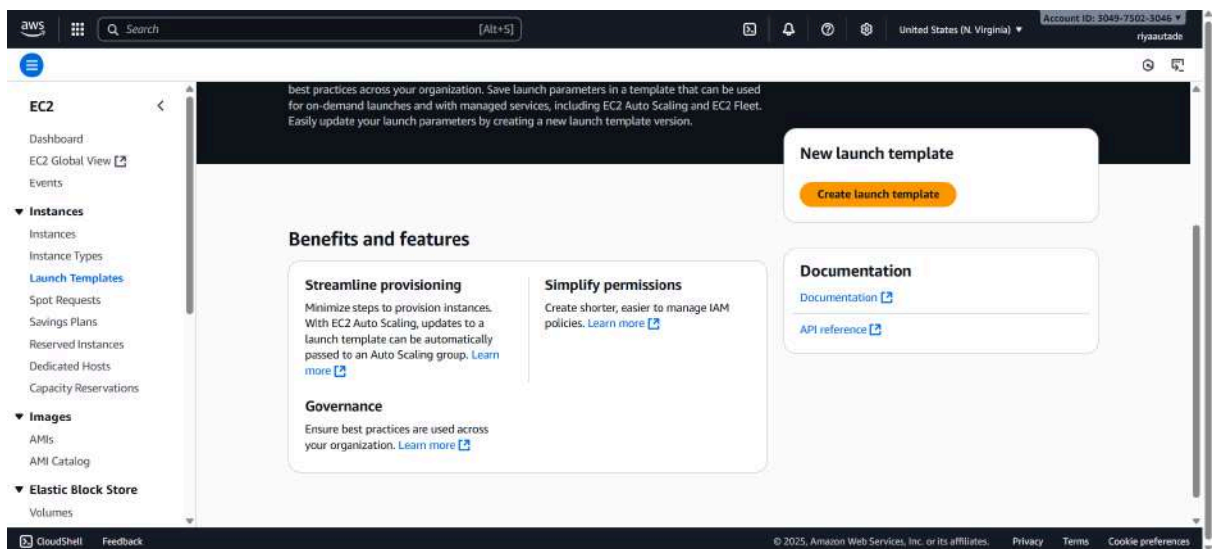
Health check path: /

Advanced health check settings:

- Health check port:** Traffic port (selected), Override (unselected)
- Healthy threshold:** 5 consecutive health check successes



5. Launch templates



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EC2 > Launch templates > Create launch template

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - required

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '"', 'gfi'.

Template version description

Max 255 chars.

Auto Scaling guidance

Select this if you intend to use this template with EC2 Auto Scaling

☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

Template tags

Source template

Summary

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-08982f1c5b95d976

Virtual server type (instance type)
-

Firewall (security group)
-

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Create launch template](#)

Launch template contents

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EC2 > Launch templates > Create launch template

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

Application and OS Images (Amazon Machine Image) - required

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Recents

My AMIs

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-08982f1c5b95d976 (64-bit x86), uefi-preferred / ami-039f81f5ce6752b10 (64-bit ARM), uefi
Virtualization: hvm ENX enabled: true Root device type: sbs

Free tier eligible

Description

Summary

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-08982f1c5b95d976

Virtual server type (instance type)
-

Firewall (security group)
-

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Create launch template](#)

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EC2 > Launch templates > Create launch template

Instance type

Instance type

t3.micro Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand Linux base pricing: 0.0104 USD per Hour

On-Demand RHEL base pricing: 0.0392 USD per Hour On-Demand Windows base pricing: 0.0195 USD per Hour

[Additional costs apply for AMIs with pre-installed software](#)

☐ All generations [Compare instance types](#)

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

Don't include in launch template [Create new key pair](#)

Network settings

Subnet

Don't include in launch template [Create new subnet](#)

When you specify a subnet, a network interface is automatically added to your template.

Summary

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-08982f1c5b95d976

Virtual server type (instance type)
t3.micro

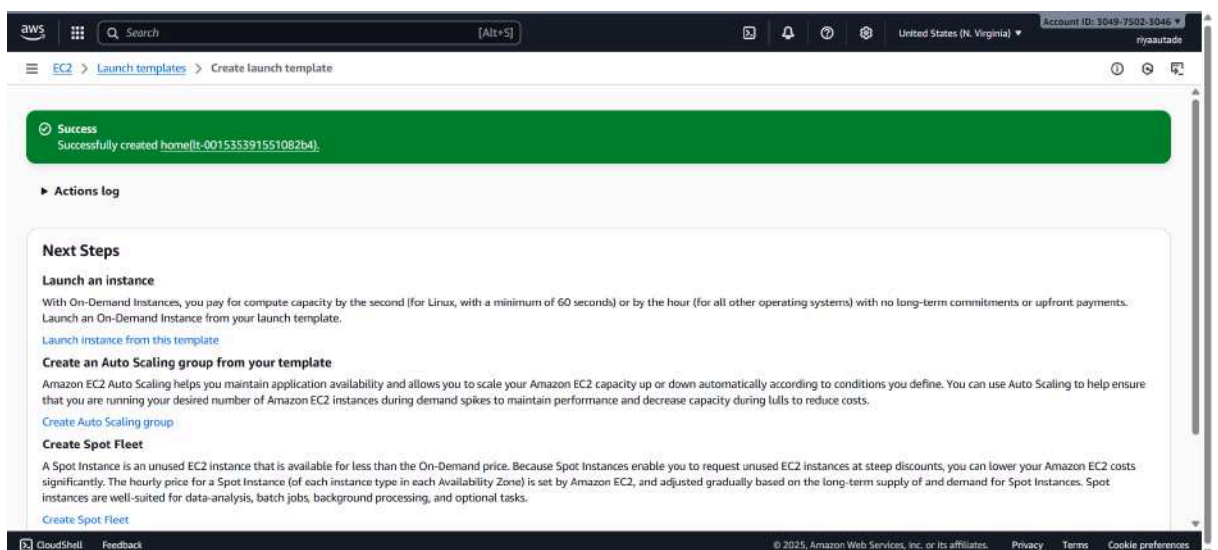
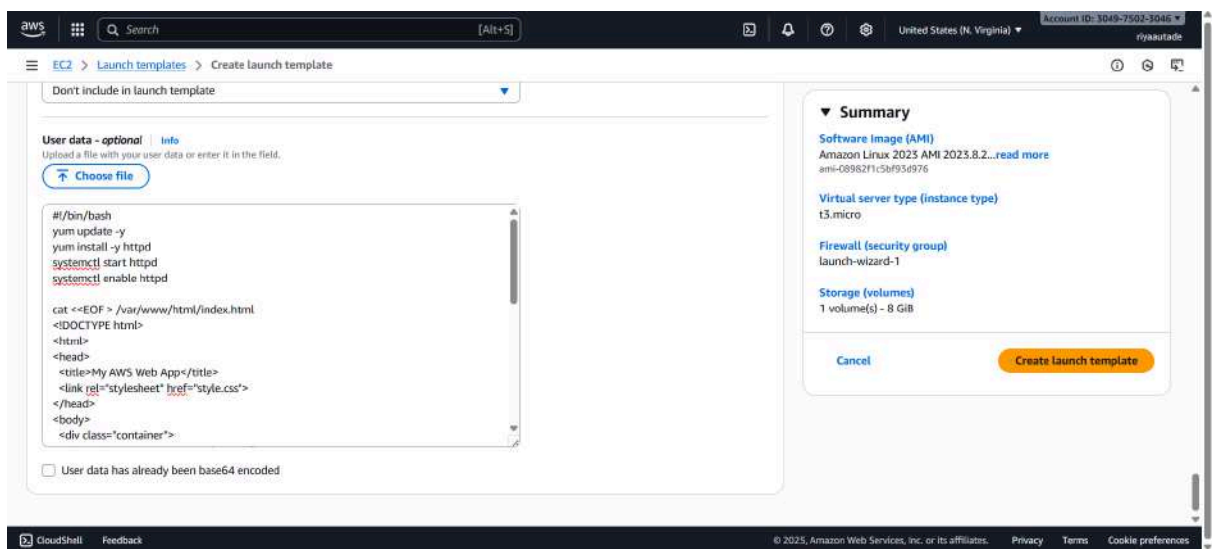
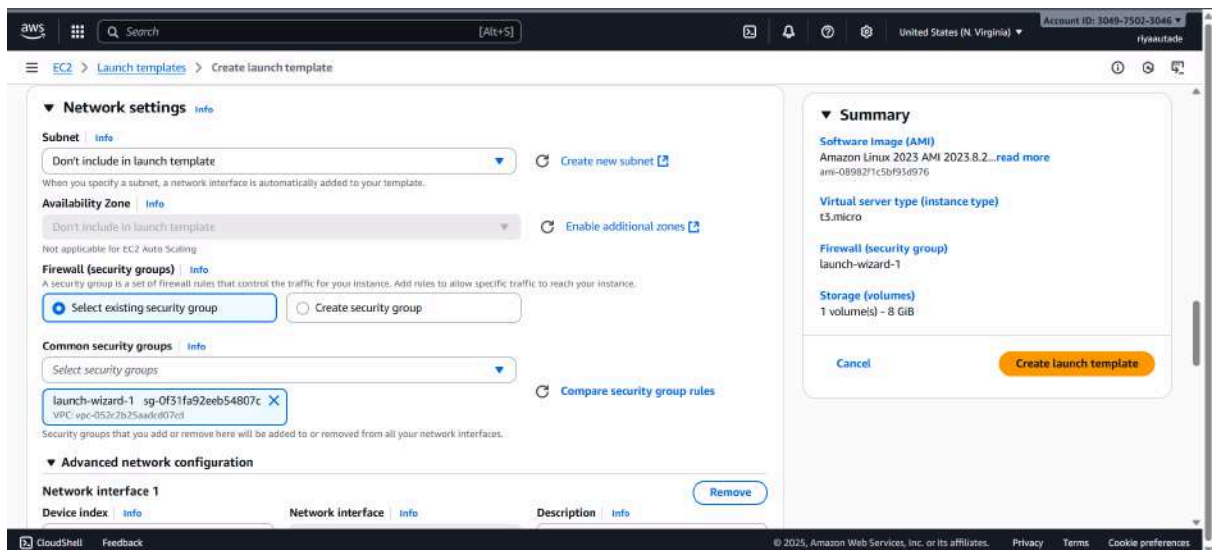
Firewall (security group)
launch-wizard-1

Storage (volumes)
1 volume(s) - 8 GiB

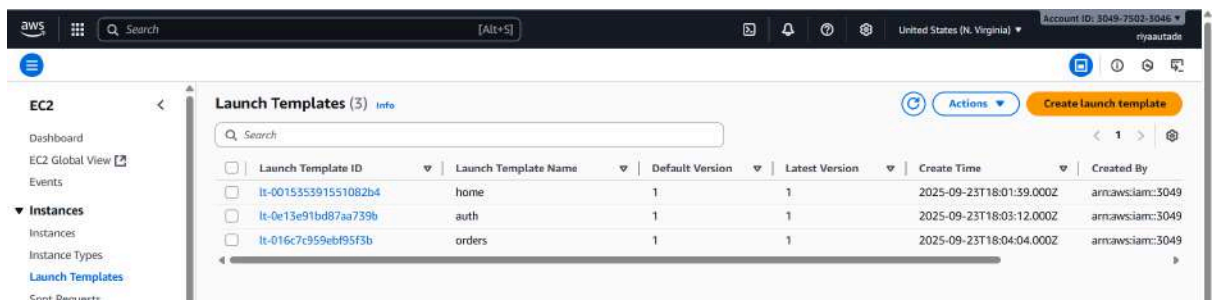
[Cancel](#) [Create launch template](#)

CloudShell Feedback

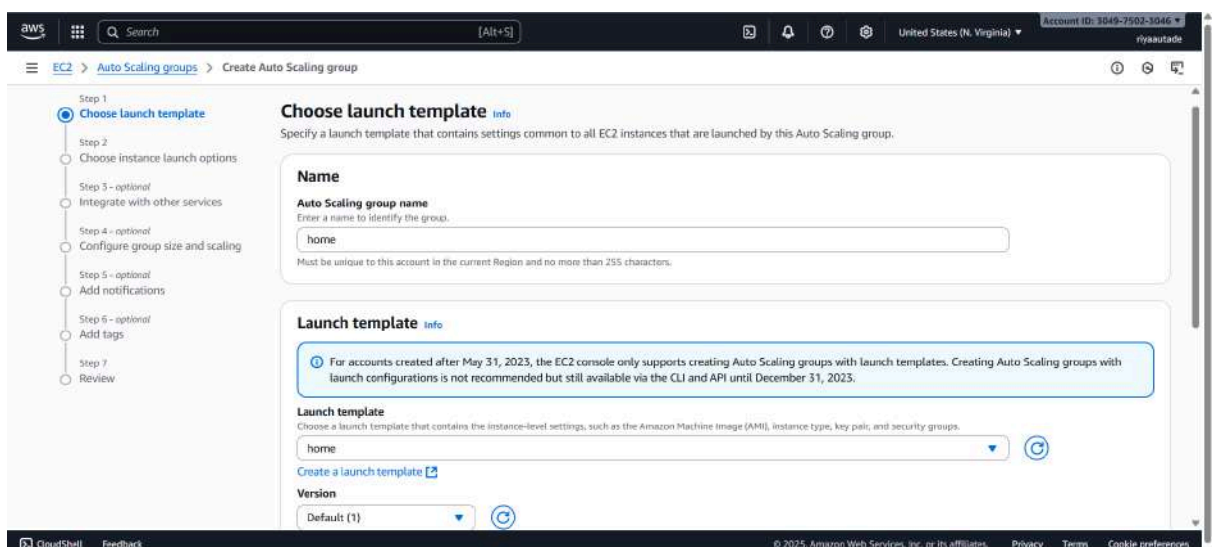
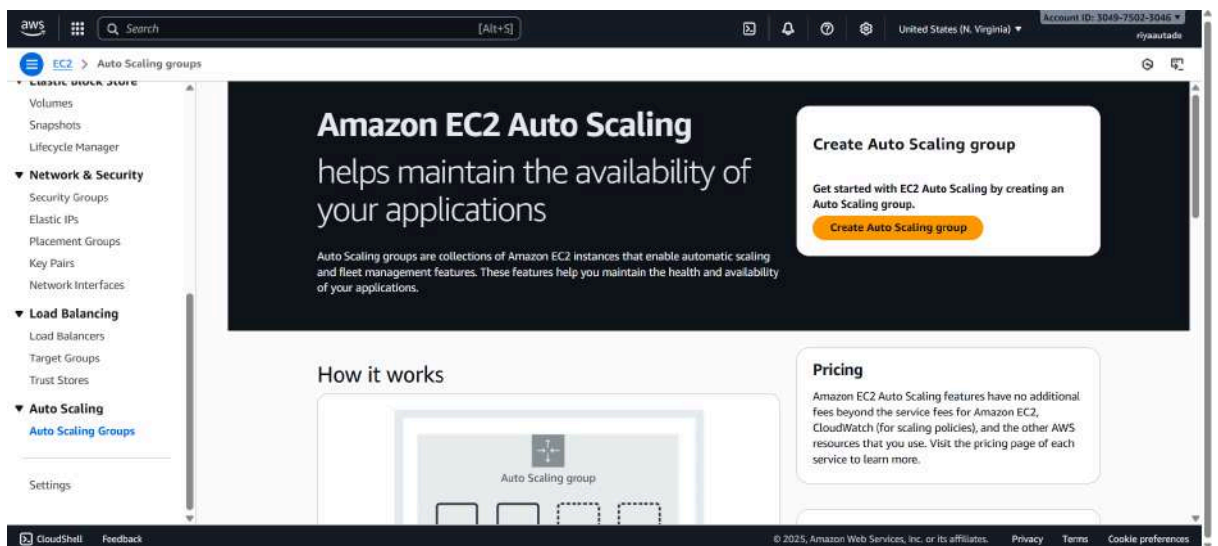
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Repeat for auth and orders



6. Auto scaling groups



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EC2 > Auto Scaling groups > Create Auto Scaling group

Review

Network info

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-052c2b25aadcd07cd
172.31.0.0/16 Default

Create a VPC

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

use1-az4 (us-east-1b) | subnet-02bd16c31dba58e62
172.31.16.0/20 Default

Create a subnet

Availability Zone distribution - new
Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

☒ **Balanced best effort**
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

☐ **Balanced only**
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

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EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1
Choose launch template

Step 2
Choose instance launch options

Step 3 - optional
Integrate with other services

Step 4 - optional
Configure group size and scaling

Step 5 - optional
Add notifications

Step 6 - optional
Add tags

Step 7
Review

Integrate with other services - optional info

Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing info

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

☐ **No load balancer**
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☒ **Attach to an existing load balancer**
Choose from your existing load balancers.

☐ **Attach to a new load balancer**
Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to an existing load balancer

Select the load balancers that you want to attach to your Auto Scaling group.

☒ **Choose from your load balancer target groups**
This option allows you to attach Application, Network, or Gateway Load Balancers.

☐ **Choose from Classic Load Balancers**

Existing load balancer target groups

Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

home | HTTP
Application Load Balancer ALB

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aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 3049-7502-3046

ryaautade

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1
Choose launch template

Step 2
Choose instance launch options

Step 3 - optional
Integrate with other services

Step 4 - optional
Configure group size and scaling

Step 5 - optional
Add notifications

Step 6 - optional
Add tags

Step 7
Review

Configure group size and scaling - optional info

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size info

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity

Specify your group size.

1

Scaling info

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

1

Equal or less than desired capacity

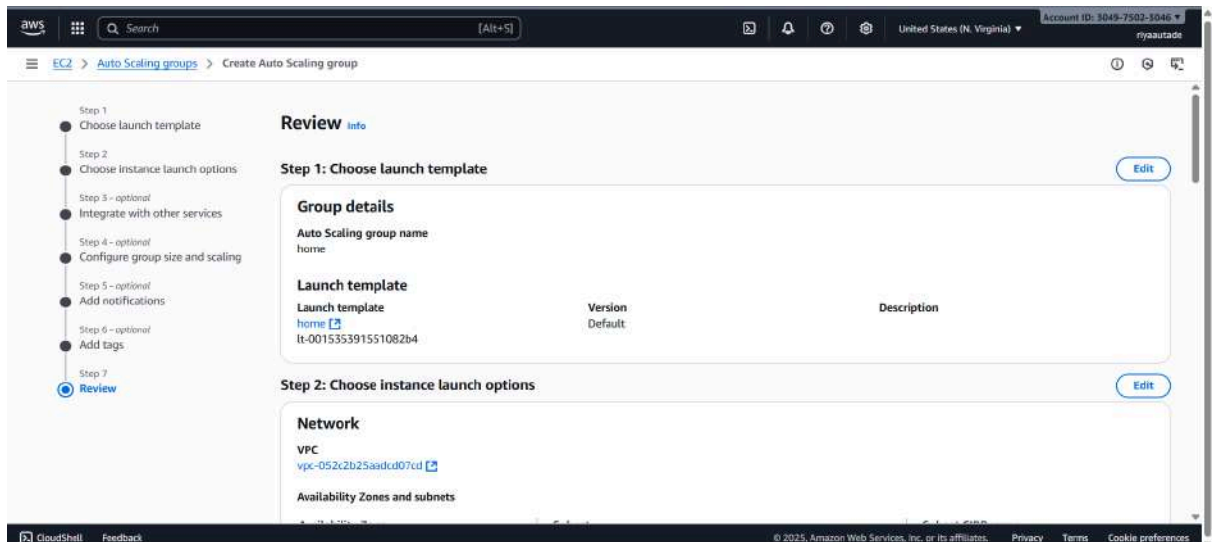
Max desired capacity

2

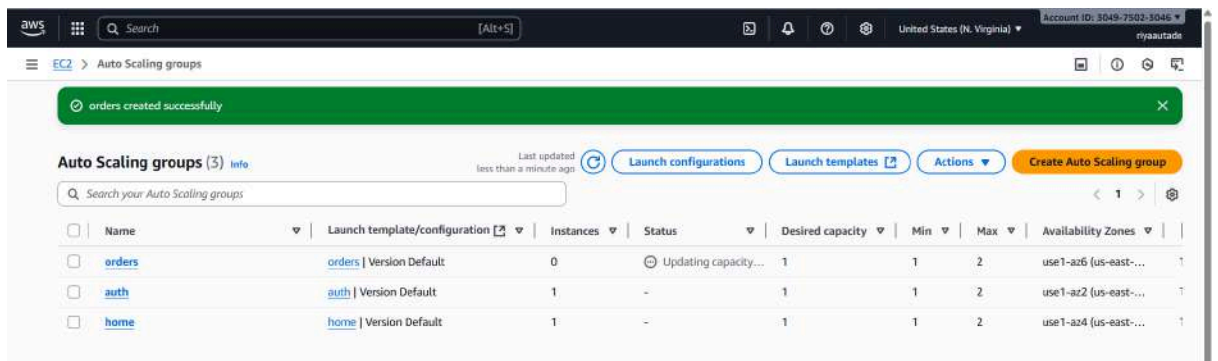
Equal or greater than desired capacity

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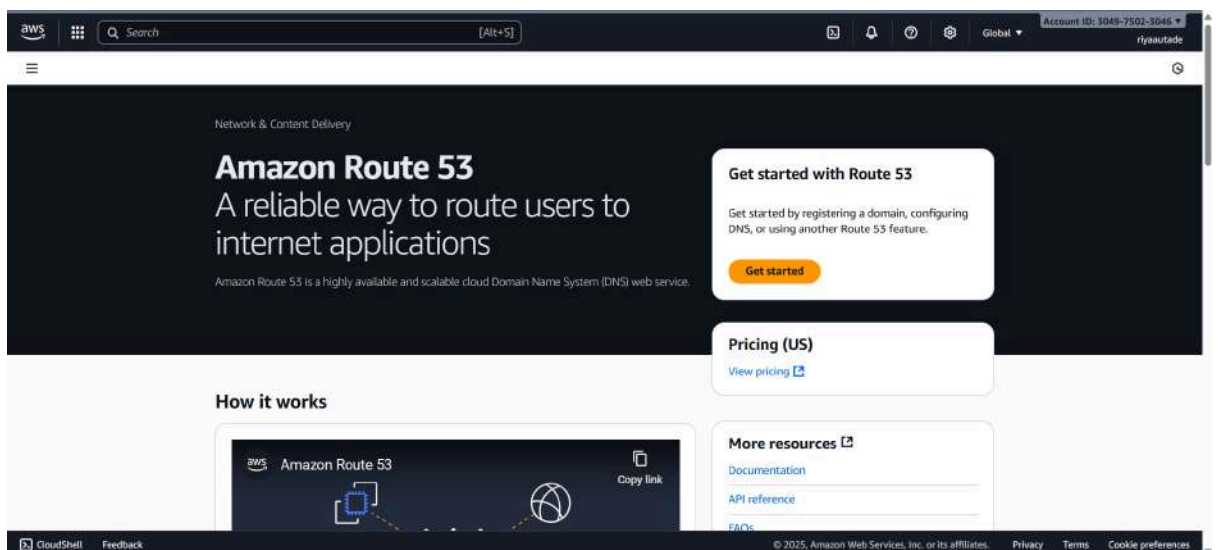
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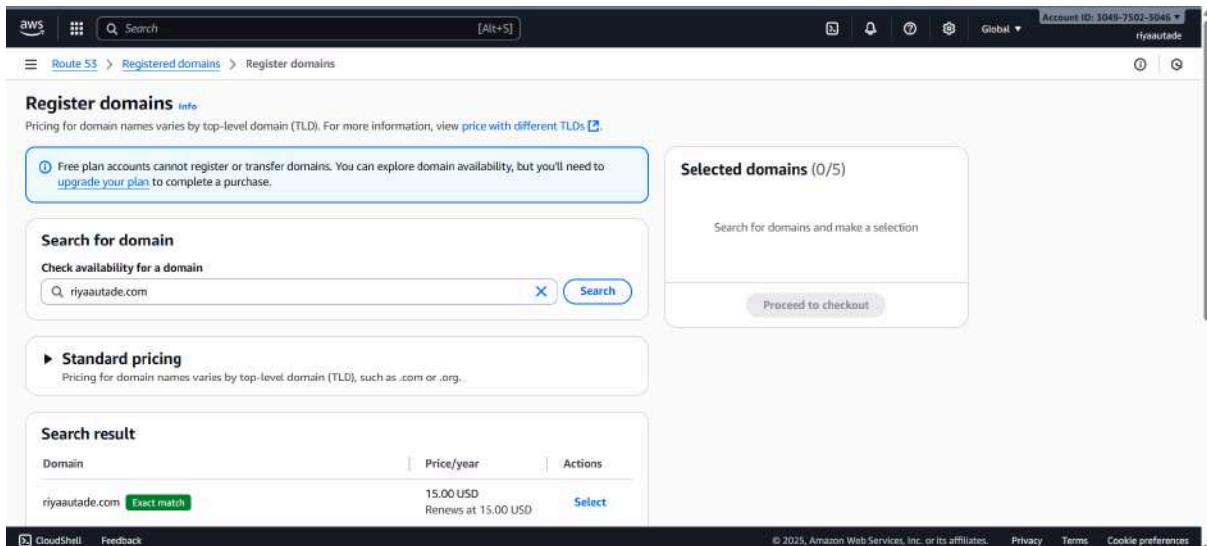


Repeat for auth and orders



7. Route 53





RESULT:

